

The Risk and Term Structure of Interest Rates

Chapter 6

Econ 351 – Monetary Economics

University of Tennessee, Knoxville

August 24, 2026

Preview

- ▶ This chapter explains why bonds with different characteristics have different interest rates.
- ▶ We study the risk structure and the term structure of interest rates.

Risk structure: bonds with the *same maturity* differ in yield because of default risk, liquidity, and taxes.

Term structure: bonds with the *same risk* differ in yield because of time to maturity (the yield curve).

Outline

Introduction

Risk Structure of Interest Rates

Term Structure of Interest Rates

Objective

- ▶ What interest rate are we studying?
- ▶ We examine both:
 - ▶ An overall (common) component shared across interest rates.
 - ▶ Individual (specific) components of particular interest rates.
- ▶ Interest rates are related to one another, but some factors cause their correlations to be less than perfect.

Effects of Monetary Policy

Effects on the Macroeconomy

- ▶ Lower policy interest rates → higher investment → economic expansion.
- ▶ Lower yields on assets → higher asset prices.

When Monetary Policy Becomes Less Effective

- ▶ Correlations among interest rates may weaken.



10-year minus 3-month Treasury yield: long and short rates do not always move together.

Source: FRED, Federal Reserve Bank of St. Louis

Factors Affecting the Level of Interest Rates

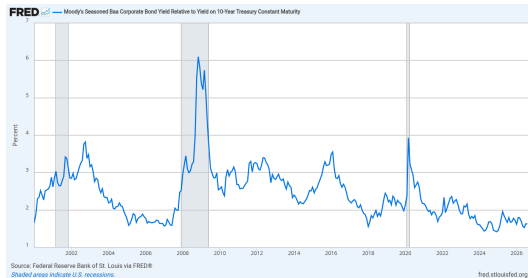
Holding Other Factors Constant

► Risk

- Riskier assets require higher interest rates.
- $r = r_f + \text{risk premium}$

► Maturity

- Longer maturity → higher interest rate.



Baa corporate yield minus 10-year Treasury yield: the risk premium is always positive and spikes in crises.

Source: FRED, Federal Reserve Bank of St. Louis

Decomposing Differences in Interest Rates

How Do We Analyze Differences in Interest Rates?

- ▶ Use a simplified framework: hold other factors constant.
- ▶ Study maturity differences alone:
 - ▶ Same risk, different maturities → term structure of interest rates.
- ▶ Study risk differences alone:
 - ▶ Same maturity, different risk → risk structure of interest rates.

When Risk and Maturity Both Differ

- ▶ First decompose, then analyze each component separately.

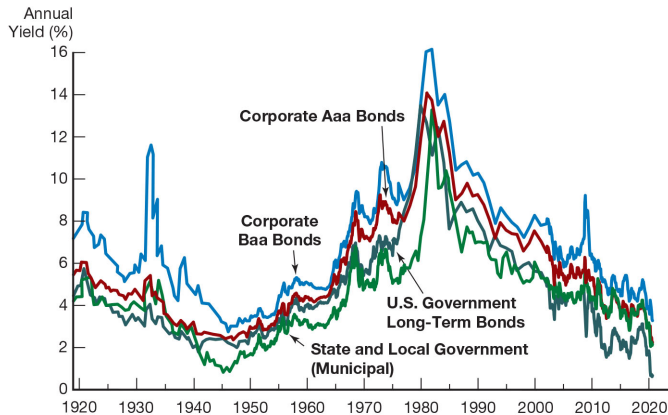
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Yields on Bonds with the Same Maturity



Long-term bond yields, 1919–2020: corporate Baa, corporate Aaa, U.S. government, and municipal bonds.

Source: Mishkin, The Economics of Money, Banking and Financial Markets

Risk Structure: Three Factors

Bonds with the same maturity have different interest rates because of

- ▶ Default risk
- ▶ Liquidity
- ▶ Tax considerations

Default occurs when an issuer is unwilling or unable to make promised interest payments or to repay principal at maturity.

A **default-free bond** has essentially no default risk.

Default Risk

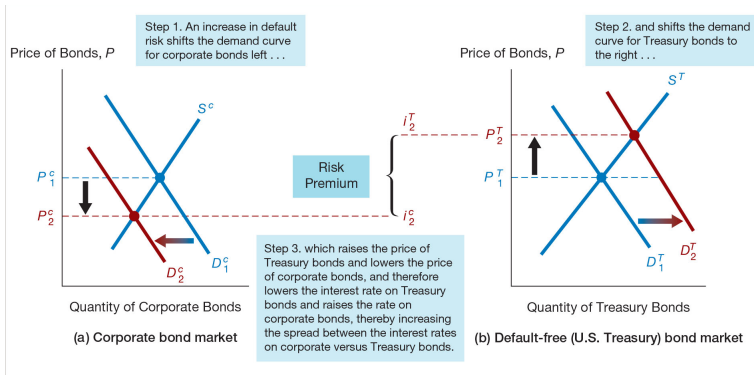
Do Government Bonds Have Default Risk?

- ▶ It depends on whether the government controls its own currency.
- ▶ **Foreign-currency debt:** default risk can be significant.
- ▶ **Local-currency debt:** default risk is typically low when the issuer can tax and issue money.
 - ▶ Example: euro-area members can face sovereign default risk (e.g., the Greek debt crisis).
 - ▶ Example: U.S. Treasury default risk is extremely low, but political standoffs (debt ceiling) can raise concerns.

Risk Premium (Default Spread)

- ▶ The yield difference between risky bonds and default-free bonds of the same maturity.

Response to an Increase in Default Risk



Higher default risk on corporate bonds shifts demand from corporate to Treasury bonds; the risk premium rises.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Credit Ratings

Credit Rating Agencies

- ▶ Moody's
- ▶ Standard & Poor's (S&P)
- ▶ Fitch
- ▶ **Investment-grade:** High credit quality, low default risk.
- ▶ **Junk (speculative):** Low credit quality, high default risk.

Moody's	Rating Agency S&P	Fitch	Definitions
Aaa	AAA	AAA	Prime Maximum Safety
Aa1	AA+	AA+	High Grade High Quality
Aa2	AA	AA	
Aa3	AA-	AA-	
A1	A+	A+	Upper Medium Grade
A2	A	A	
A3	A-	A-	
Baa1	BBB+	BBB+	Lower Medium Grade
Baa2	BBB	BBB	
Baa3	BBB-	BBB-	
Ba1	BB+	BB+	Noninvestment Grade
Ba2	BB	BB	Speculative
Ba3	BB-	BB-	
B1	B+	B+	Highly Speculative
B2	B	B	
B3	B-	B-	
Caa1	CCC+	CCC	Substantial Risk
Caa2	CCC	—	In Poor Standing
Caa3	CCC-	—	
Ca	—	—	Extremely Speculative
C	—	—	May Be in Default
—	—	D	Default

Bond ratings by Moody's, S&P, and Fitch.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Michael Milken and Junk Bonds

Two Stages in the Development of Junk Bonds

- ▶ Seeking high returns by investing in existing junk bonds.
- ▶ Creating new junk bonds to finance corporate takeovers.

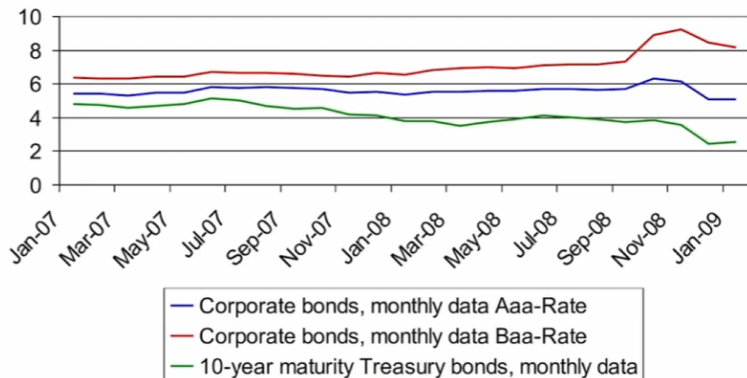
[Michael Milken \(Wikipedia\)](#)



Michael Milken, the “junk bond king” of the 1980s, in 2026.

Wikimedia Commons, Xuthoria, CC BY-SA 4.0

Corporate Bond Risk Premium During the Crisis



Corporate bond and Treasury yields during the 2007–2009 financial crisis.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

- ▶ The Baa–Treasury spread widens sharply.
- ▶ Default risk increases during crises.

The Coronavirus Pandemic and the Baa–Treasury Spread

Shock to the Economy

- ▶ March 2020: Lockdowns sharply reduced economic activity.
- ▶ Corporate revenues fell, increasing default risk.

Market Response

- ▶ Investors demanded safer assets (flight to quality).
- ▶ Demand for Baa bonds shifted left.
- ▶ The Baa–Treasury spread widened significantly.



Baa–Treasury spread since 2000: spikes in 2008–09 and March 2020.

Source: *FRED, Federal Reserve Bank of St. Louis*

Liquidity and Income Tax Considerations

Liquidity

Liquidity measures how easily an asset can be converted into cash without affecting its price. Determinants:

- ▶ **Transaction costs:** The cost of selling the asset.
- ▶ **Market depth:** The number of buyers and sellers in the market.

Income Tax Considerations

- ▶ Interest payments on municipal bonds are **exempt from federal income taxes**.
- ▶ The tax advantage makes municipal bonds attractive despite lower pre-tax returns.

Effects of the 2017 Tax Cuts on Bond Interest Rates

The 2017 Tax Cuts and Jobs Act reduced the top marginal income tax rate from 39.6% to 37%.

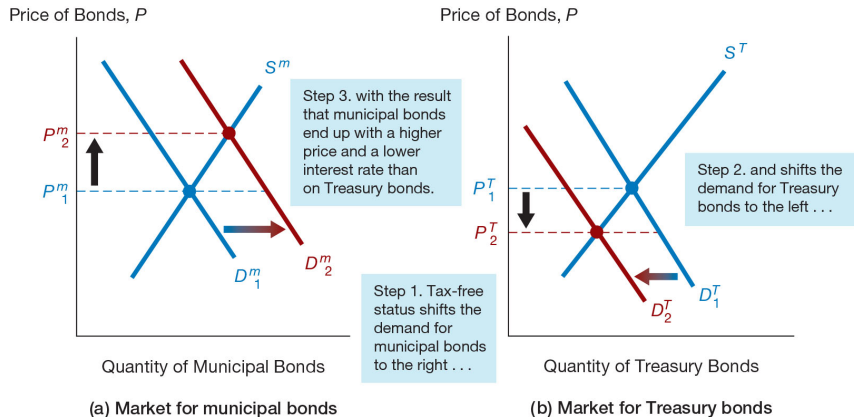
Economic Mechanism

- ▶ Municipal bond interest is tax-exempt.
- ▶ Lower tax rates reduce the tax advantage of municipal bonds.
- ▶ After-tax returns change across bonds.

Market Impact

- ▶ **Municipal bonds:** lower tax benefit → demand shifts left → price falls, interest rate rises.
- ▶ **Treasury bonds:** relatively more attractive → demand shifts right → price rises, interest rate falls.

Municipal and Treasury Bond Markets



Interest rates on municipal and Treasury bonds: the tax exemption shifts demand toward municipals.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Risk Structure of Interest Rates: Summary

How Do Risky and Safe Interest Rates Move?

- ▶ Sometimes they move **together**.
- ▶ Sometimes they move in **different directions**.

When the Risk Premium Is Stable

- ▶ Interest rates tend to rise and fall together.
- ▶ The spread between risky and Treasury bonds stays about the same.

When the Risk Premium Changes

- ▶ Risky and safe rates diverge.
- ▶ The spread widens or narrows.

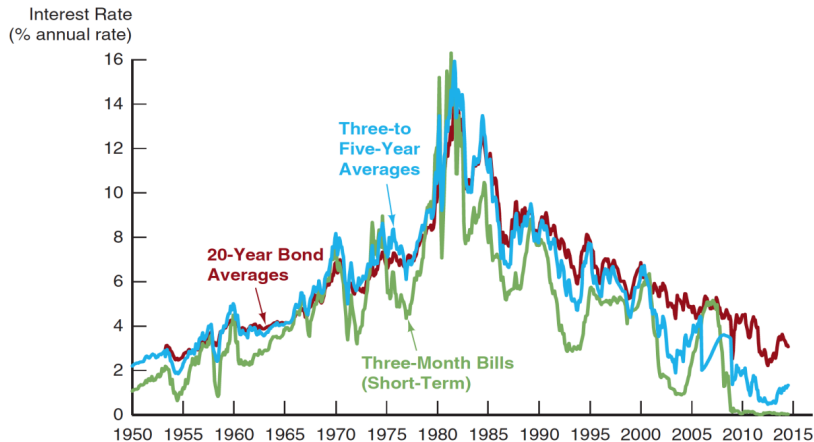
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Term Structure of Interest Rates

Interest Rates on Bonds with Different Maturities



Movements over time of interest rates on U.S. government bonds with different maturities.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

The Yield Curve

Bonds with identical risk, liquidity, and tax characteristics may have different interest rates because the time remaining to maturity is different.

Yield Curve

- ▶ A graph showing interest rates on bonds with different maturities (holding risk, liquidity, and tax treatment constant).

Common Shapes

- ▶ **Upward-sloping (normal):** long-term rates $>$ short-term rates
- ▶ **Flat:** long-term rates $\approx$ short-term rates
- ▶ **Inverted:** long-term rates $<$ short-term rates

Three Facts a Theory Must Explain

What Must a Theory Explain?

1. Interest rates of different maturities tend to move together.
2. When short-term rates are low, the yield curve is usually upward-sloping; when short-term rates are high, it is more likely to flatten or invert.
3. Most of the time, yield curves slope upward.



Yield-curve slope (10-year minus 3-month Treasury):
usually positive, inverted before recessions.

Source: FRED, Federal Reserve Bank of St. Louis

Three Theories of the Term Structure

1. Expectations Theory

- ▶ Perfect substitutes $\Rightarrow$ rates move together $\Rightarrow$ cannot explain upward slope

2. Segmented Markets Theory

- ▶ No substitution $\Rightarrow$ explains upward slope $\Rightarrow$ cannot explain co-movement

3. Liquidity Premium Theory

- ▶ Imperfect substitutes $\Rightarrow$ adds maturity premium $\Rightarrow$ explains all three facts

Expectations Theory

- ▶ The yield on a long-term bond equals the average of expected future short-term rates over its life.
- ▶ Investors have no maturity preference.
- ▶ Bonds of different maturities are **perfect substitutes**.
- ▶ Investors hold the bond with the highest expected return.

An example: the one-year rate is 6% today and is expected to be 8% next year. Buying two one-year bonds in a row yields on average

$$\frac{6\% + 8\%}{2} = 7\%,$$

so the two-year rate must be 7% for you to be willing to buy the two-year bond.

Expectations Theory: Derivation (1/2)

Consider a \$1 investment. Notation:

- ▶ i_t = one-period interest rate today
- ▶ i_{t+1}^e = expected one-period rate next period
- ▶ i_{2t} = two-period interest rate today

Expected return over two periods from the two-period bond:

$$(1 + i_{2t})(1 + i_{2t}) - 1 = 2i_{2t} + (i_{2t})^2 \approx 2i_{2t}$$

since $(i_{2t})^2$ is very small.

Expected return from two one-period bonds:

$$(1 + i_t)(1 + i_{t+1}^e) - 1 = i_t + i_{t+1}^e + i_t i_{t+1}^e \approx i_t + i_{t+1}^e$$

Expectations Theory: Derivation (2/2)

Both bonds will be held only if the expected returns are equal:

$$2i_{2t} = i_t + i_{t+1}^e \quad \implies \quad i_{2t} = \frac{i_t + i_{t+1}^e}{2}.$$

For bonds with longer maturities:

$$i_{nt} = \frac{i_t + i_{t+1}^e + i_{t+2}^e + \cdots + i_{t+(n-1)}^e}{n}.$$

The n -period rate is the average of the current one-period rate and the expected one-period rates over the next $n - 1$ periods.

Practice Problems: Expectations Theory

Problem 1

Suppose the expected one-year interest rates for the next five years are:

3%, 4%, 5%, 5%, 4%

What is the expected two-year interest rate two years from now?

Problem 2

Suppose expected one-year rates over the next ten years are:

- ▶ Years 1–3: 4%
- ▶ Years 4–7: 5%
- ▶ Years 8–10: 6%

If an investor buys a 5-year bond at the end of year 5, what is the expected yield to maturity?

What Expectations Theory Explains

- ▶ Why the term structure of interest rates changes at different times.
- ▶ Why interest rates on bonds with different maturities move together over time (Fact 1).
- ▶ Why yield curves tend to slope up when short-term rates are low and slope down when short-term rates are high (Fact 2).
- ▶ It cannot explain why yield curves usually slope upward (Fact 3).

Segmented Markets Theory

- ▶ Bonds of different maturities are **not substitutes**.
- ▶ Each maturity has its **own supply and demand**, which determines its interest rate.
- ▶ Investors have **preferred maturities**.
- ▶ If investors prefer short-term bonds (lower interest-rate risk), long-term bonds must offer **higher yields** to attract buyers (Fact 3).

Because each maturity is a separate market, the theory cannot explain why rates of different maturities move together (Fact 1) or why the slope depends on the level of short rates (Fact 2).

Liquidity Premium Theory

- ▶ The long-term interest rate equals the average of expected future short-term rates plus a liquidity premium:

$$i_{nt} = \frac{i_t + i_{t+1}^e + i_{t+2}^e + \cdots + i_{t+(n-1)}^e}{n} + l_{nt}$$

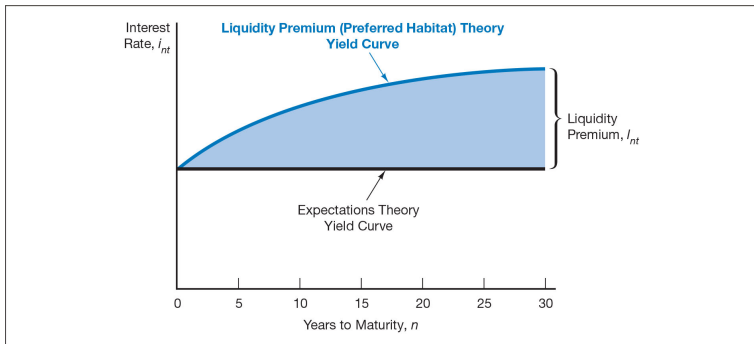
- ▶ l_{nt} is the liquidity premium for the n -period bond at time t ; it is always positive and rises with the term to maturity.
- ▶ The liquidity premium compensates investors for **interest-rate risk** and depends on supply and demand.
- ▶ Bonds of different maturities are **imperfect substitutes**.

Preferred Habitat Theory

- ▶ Investors have a preference for bonds of one maturity over another.
- ▶ They will be willing to buy bonds of different maturities only if they earn a somewhat higher expected return.
- ▶ Investors are likely to prefer short-term bonds over longer-term bonds.

Preferred habitat and liquidity premium theories lead to the same conclusion: long-term rates equal the average of expected short-term rates plus a positive term premium that rises with maturity.

Liquidity Premium vs. Expectations Theory



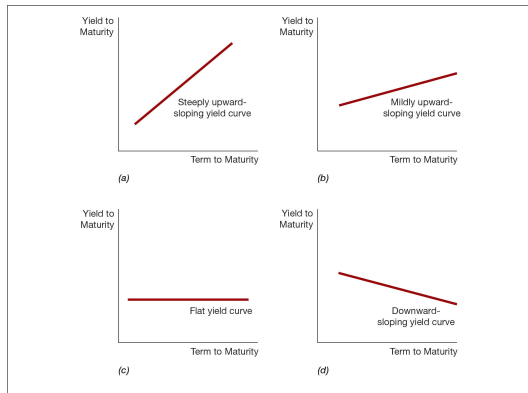
The liquidity premium (preferred habitat) yield curve lies above the expectations-theory curve by the premium l_{nt} .

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

How Liquidity Premium Theory Explains the Three Facts

- ▶ **Rates move together:** because long-term rates reflect the average of expected future short-term rates.
- ▶ **Upward slope when short rates are low:** the liquidity premium dominates.
- ▶ **Inverted curve when short rates are high:** markets expect future short-term rates to fall.
- ▶ **Yield curves usually slope upward:** the liquidity premium increases with maturity.

Yield Curves and Expected Future Short-Term Rates



Liquidity premium (preferred habitat) theory. (a) Steeply rising: short rates expected to rise. (b) Moderately steep: short rates expected to stay about the same. (c) Flat: short rates expected to fall moderately. (d) Inverted: short rates expected to fall sharply.

Source: Mishkin, *The Economics of Money, Banking and Financial Markets*

Chapter summary

- ▶ **Risk structure:** bonds of the same maturity differ in yield because of default risk (risk premium), liquidity, and tax treatment.
- ▶ Risk premiums widen in crises (2008–09, March 2020) as investors flee to Treasuries.
- ▶ **Term structure:** the yield curve must explain three facts: rates move together, the slope depends on the level of short rates, and curves usually slope upward.
- ▶ Expectations theory explains Facts 1 and 2; segmented markets explains Fact 3.
- ▶ Liquidity premium (preferred habitat) theory combines them: long rate = average expected short rates + a positive, maturity-increasing premium, explaining all three facts.